

EVORA AP, Portugal

WMO: 085580

Lat: 38.5367N

Lon: 7.8881W

Elev: 806

StdP: 14.27

Time Zone: 0.00 (EUW)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	34.0	36.2	22.3	17.3	45.6	26.7	21.2	45.9	24.1	51.5	21.5	51.8	4.2	0	0.464

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	29.8	99.7	67.4	96.5	66.9	93.3	65.9	70.1	91.5	69.0	89.3	67.9	87.0	6.4	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.0	95.4	71.1	63.7	90.9	70.6	62.5	87.0	70.3	34.5	91.7	33.5	89.7	32.5	87.1	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.4	17.7	15.6	29.4	105.0	2.4	3.5	27.6	107.5	26.2	109.6	24.8	111.5	23.1	114.0		
			27.7	72.0	2.6	1.4	25.8	73.0	24.3	73.8	22.8	74.6	20.9	75.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.7	48.9	50.6	54.7	57.7	63.9	71.6	75.5	76.4	71.4	64.4	54.6	50.1
	DBStd	11.09	4.38	4.47	4.91	5.46	6.37	6.39	5.23	5.09	5.01	5.39	5.02	4.38
	HDD50	202	73	44	13	4	0	0	0	0	0	0	14	54
	HDD65	2418	500	402	319	228	98	14	1	0	5	76	312	463
	CDD50	4479	38	61	160	234	433	647	790	818	643	446	153	56
	CDD65	1219	0	0	1	8	65	211	326	353	198	56	1	0
	CDH74	14883	0	0	24	148	907	2720	4100	4384	2114	483	3	0
	CDH80	7479	0	0	0	20	330	1376	2255	2427	931	140	0	0
Wind	WSAvg	6.7	6.2	6.7	7.1	7.2	6.6	7.4	7.9	7.1	6.1	6.0	6.3	6.0
Precipitation	PrecAvg	25.1	3.5	3.1	2.1	2.2	1.9	1.0	0.3	0.2	1.1	2.8	3.2	3.5
	PrecMax	40.9	13.5	8.2	8.2	6.9	4.8	4.1	2.9	2.9	3.0	8.5	9.6	9.8
	PrecMin	14.4	0.0	0.2	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.1	0.2	0.2
	PrecStd	7.0	3.1	2.5	1.9	1.5	1.4	1.1	0.6	0.5	0.9	2.4	2.6	2.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.4	67.8	77.1	82.7	92.9	101.2	103.7	103.3	97.6	89.7	73.4	65.1
		MCWB	54.8	56.5	58.3	62.2	64.7	67.5	67.9	68.8	66.6	62.6	60.4	56.6
	2%	DB	60.5	64.4	73.4	78.3	88.0	96.6	99.7	99.6	93.1	83.7	69.1	61.6
		MCWB	53.3	54.2	57.1	61.1	63.4	66.8	66.9	67.9	65.0	61.9	58.4	57.1
	5%	DB	58.4	61.7	69.7	74.4	83.4	92.7	95.9	96.2	89.6	79.2	65.7	59.8
		MCWB	52.8	53.7	56.4	59.6	62.4	65.9	66.4	67.4	64.3	61.8	58.0	55.7
	10%	DB	56.5	59.1	65.7	70.1	78.9	88.4	92.0	92.4	85.7	75.1	63.1	57.7
		MCWB	52.3	52.7	55.4	58.0	61.5	65.0	65.9	66.1	63.8	60.8	57.0	54.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.0	59.1	62.1	65.7	68.4	70.8	71.1	71.9	69.8	68.3	64.9	61.0
		MCDB	61.0	64.1	72.7	79.1	84.8	92.3	92.0	95.1	90.8	75.7	67.7	61.8
	2%	WB	56.3	57.1	59.7	62.8	65.9	69.2	69.6	70.3	68.0	66.3	62.1	58.6
		MCDB	58.5	61.6	68.6	74.3	81.7	90.2	90.9	92.7	84.7	74.1	65.4	60.6
	5%	WB	54.8	55.5	58.0	60.5	64.2	67.6	68.3	69.0	66.8	64.8	60.1	56.8
		MCDB	56.9	59.8	66.3	71.6	78.8	87.9	89.6	90.2	81.8	72.6	63.5	58.9
	10%	WB	53.1	53.9	56.3	58.8	62.6	66.1	67.1	67.7	65.6	63.3	58.6	55.1
		MCDB	55.5	58.2	63.9	68.3	76.1	84.4	88.0	87.3	79.5	70.9	61.9	57.3
Mean Daily Temperature Range	5% DB	MDBR	15.3	16.8	19.1	20.0	23.0	27.0	29.6	29.8	24.8	19.4	15.9	14.8
		MCDBR	17.5	20.4	26.0	27.2	30.9	33.8	35.3	34.5	31.5	26.9	18.9	16.4
		MCWBR	11.5	12.1	13.2	12.6	10.8	10.0	10.0	10.1	9.7	10.0	10.5	11.0
	5% WB	MCDBR	12.5	16.4	21.4	24.2	26.5	30.7	31.3	31.1	25.5	18.7	13.8	12.8
		MCWBR	9.3	10.6	12.2	12.4	11.1	10.2	10.1	10.4	9.0	8.7	9.4	10.2
Clear-Sky Solar Irradiance	taub	0.312	0.328	0.359	0.368	0.376	0.379	0.359	0.376	0.372	0.350	0.325	0.313	
	taud	2.504	2.458	2.379	2.361	2.367	2.373	2.404	2.367	2.399	2.490	2.516	2.511	
	Ebn at Noon	276	285	285	289	287	285	290	282	277	274	268	266	
	Edh at Noon	25	30	35	38	39	38	37	37	34	28	24	23	
All-Sky Solar Radiation	RadAvg	735	1036	1389	1747	2109	2391	2481	2199	1695	1165	814	655	
	RadStd	97	128	154	165	186	133	78	75	109	102	102	76	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page